

COURSE TITLE : FIBRES AND COMPOSITES
COURSE CODE : 5091
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 65/5
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Natural fibres	16
2	Synthetic fibres	16
3	Composites and its constituents	16
4	Composite products	17
TOTAL		65

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the manufacture ,properties and applications various natural fibres
2	2	To understand the manufacture ,properties and applications various synthetic fibres
3	3	To know the classification of composites and its constituents
4	4	To understand the processing of different composite products

SPECIFIC OUTCOME

MODULE I NATURAL FIBRES

1.1.0 To understand the manufacture, properties and applications various natural fibres

- 1.1.1. Define the terms fibre,Yarn, cord,strand, Tex,Denier,Tenacity,Warp,Weft and Twist
- 1.1.2. List different Natural fibers and classify them into vegetable,animal and mineral fibres.
- 1.1.3. Explain the sources,processing ,properties and applications of cotton,Linen(flax) ,Jute,Sisal and coir fibres.
- 1.1.4. Explain the sources,processing ,properties and applications of silk and wool
- 1.1.5. Explain the processing ,properties and applications of Asbestose fibre

MODULE II SYNTHETIC FIBRES

2.1.0 To understand the manufacture, properties and applications of various synthetic fibres

- 2.1.1. Explain the melt spinning and solution spinning processes for the conversion of polymer into fibres
- 2.1.2. Describe the manufacture, properties and applications of Viscose Rayon and Cellulose acetate
- 2.1.3. Explain the Raw polymer, fibre formation, properties and applications of Nylon, Polyester, Acrylic, PP and Aramid fibres
- 2.1.4. Explain the manufacture, properties and applications of Glass fibre
- 2.1.5. Give the different forms and grades of commercially available forms of Glass fibre with its characteristics
- 2.1.6. Describe the manufacture, properties and applications of carbon fibre

MODULE III COMPOSITES AND ITS CONSTITUENTS

3.1.0 To know the classification of composites and its constituents

- 3.1.1. Define Polymer composites
- 3.1.2. Classify the composites according to polymer matrix and reinforcement
- 3.1.3. Explain the function and performance characteristics of constituents of composites-resins, reinforcements, catalysts, accelerators and fillers with examples
- 3.1.4. State the advantages and disadvantages of composites over conventional materials
- 3.1.5. Explain the Rule of mixtures

MODULE IV COMPOSITE PRODUCTS

4.1.0 To understand the processing of different composite products

- 4.1.1. List the various processing techniques for the manufacture of composite products
- 4.1.2. Explain hand-layup, spray layup, pultrusion, vacuum bag moulding, pressure bag moulding, centrifugal casting, filament winding, reaction injection moulding.
- 4.1.3. Differentiate between SMC and DMC
- 4.1.4. Explain the design factors such as selection of resins and ingredients, process, section thickness and corners with respect to manufacture of composite products.
- 4.1.5. State the advantages and disadvantages of FRP mould materials-POP, wood, steel, and FRP
- 4.1.6. Describe the method of cost calculation of composite products
- 4.1.7. Explain the manufacture of Boat hull, Helmet, Roofing sheet, Bath Tub, Water tank Automobile parts-bumper and dash board with special emphasis on raw material selection and process.
- 4.1.8. Define polymer hybrid composites and nano composites
- 4.1.9. Explain the manufacture, advantages and applications of polymer hybrid composites

CONTENT DETAILS

MODULE I

Define—fibre, yarn, cord, strand, tex, denier, tenacity, warp, weft and twist. Natural fibres-classification-vegetable, animal and mineral fibres-examples. Processing-properties-applications-Cotton, Linen, Jute, Sisal , Coir, Silk, Wool and Asbestose Fibres.

MODULE II

Synthetic Fibres-Spinning process-Melt spinning, Solution spinning .Manufacture, Properties and Applications-Viscose Rayon, Cellulose Acetate. Raw polymer, Fiber formation, properties and applications -Nylons, Polyester, Acrylic, PP and Aramids. GlassfibreManufacture, Properties, applications, different forms-grades-characteristics. Carbon fibre- Manufacture, Properties, applications

MODULE III

Polymer composites-Classification-Polymer matrix and reinforcement. Composite constituents-Function, performance characteristic-Resins, Reinforcements, Catalyst, Accelerators and Fillers-Examples. Comparison with conventional materials-advantages and disadvantages. Composites-Rule of Mixtures.

MODULE IV

Composite products-Processing techniques- Hand lay up, Spray lay up, Pultrusion, Vacuum bag moulding, Pressure bag moulding, centrifugal casting, Filament winding and Reaction injection moulding. Differentiate—SMC,DMC. Design factors-selection of resins, ingredients, process, section thickness and corners. Moulds-FRP, POP, wood-advantages, disadvantages. Composite product-cost calculation. Manufacture method-Boat hull, Helmet Roofing sheet Bath tub, Water tank, automobile bumper, dash board. Define-Hybrid composite and nano composite. Polymer hybrid composite Manufacture, advantages and applications.

REFERENCES

1. Fibre science and technology - Premamoy Ghosh(Publisher Tata McGraw-Hill Education2004)
2. Man made fibres -R.W Moncrieff (Publisher –Iliffe 4th Edition)
3. FRP technology - R.G Weatherhead (Applied Science Publishers 1980)